📊 Project Title

Multi-Site Operations Performance Report (MIS Report)

📄 Project Overview

This project focuses on building an **Operational MIS Dashboard** to monitor and analyze reporting performance across multiple project sites. The objective is to track key operational metrics such as reporting compliance, delays, data quality, and issue management to support data-driven decision-making.

The dataset simulates real-world multi-site operations where different teams submit daily reports, maintain logbooks, and report issues.

🎯 Objectives

- Monitor **reporting compliance** across multiple sites
- Identify **delays in report submission**
- Analyze **data quality through error rates**
- Track and manage **operational issues**
- Provide actionable insights for **performance improvement**

🗂️ Dataset Description (Multi-Sheet Excel)

📘 1. Site_Master

Purpose: Contains basic information about each site

Columns:

- Site_ID
- Site_Name
- Region
- Project_Manager

📅 2. Daily_Reports

Purpose: Tracks daily reporting activity from each site

Columns:

- Date
- Site_ID
- Report_Submitted
- Submission_Time
- Delay_Flag
- Actual_Status *(created during data cleaning)*

📊 3. Logbook_Data

Purpose: Measures data quality of submitted reports

Columns:

- Date
- Site_ID
- Entries_Count
- Errors_Found
- Error_Rate *(created during data cleaning)*

⚠️ 4. Issues_Tracker

Purpose: Tracks operational issues across sites

Columns:

- Issue_ID
- Site_ID
- Issue_Type
- Status

🧹 Data Cleaning & Preparation (Excel)

- Standardized **date formats** and text values (Yes/No, Open/Closed)
- Handled **missing values** in submission time

- Created **Actual_Status** (On Time / Delayed / Missing)
- Validated **Delay_Flag logic** using submission time
- Removed duplicate records
- Created **Error Rate** = Errors_Found / Entries_Count
- Ensured consistency across all tables
- Maintained separate **cleaned datasets** for analysis

🔗 Data Modeling (Power BI)

- Imported all sheets into Power BI
- Established relationships using **Site_ID**
- Designed a **Star Schema**:
  - **Dimension Table:** Site_Master
  - **Fact Tables:** Daily_Reports, Logbook_Data, Issues_Tracker
- Created a **Date Table** for time-based analysis
- Ensured proper relationship direction and integrity

📊 KPI Creation (DAX Measures)

✅ Reporting Compliance %

```
Reporting Compliance % =
DIVIDE(
    COUNTROWS(FILTER(Daily_Reports, Daily_Reports[Report_Submitted] =
"Yes")),
    COUNTROWS(Daily_Reports)
)
```

✅ Delay Rate %

```
Delay Rate % =
DIVIDE(
    COUNTROWS(FILTER(Daily_Reports, Daily_Reports[Delay_Flag] = "Yes")),
    COUNTROWS(Daily_Reports)
)
```

✅ On-Time Submission %

```
On-Time % =
DIVIDE(
    COUNTROWS(FILTER(Daily_Reports, Daily_Reports[Actual_Status] = "On
```

```
Time")),
    COUNTROWS(Daily_Reports)
)
```

✅ Average Error Rate

```
Avg Error Rate =
AVERAGE(Logbook_Data[Error Rate])
```

✅ Open Issues Count

```
Open Issues =
COUNTROWS(FILTER(Issues_Tracker, Issues_Tracker[Status] = "Open"))
```

✅ Performance Score

```
Performance Score =
( [Reporting Compliance %] * 0.5 )
+ ( (1 - [Delay Rate %]) * 0.3 )
+ ( (1 - [Avg Error Rate]) * 0.2 )
```

📈 Dashboard Design (Power BI)

🏷️ Title

Multi-Site Operations Performance Dashboard

📊 KPI Section

- Reporting Compliance %
- Delay Rate %
- On-Time Submission %
- Average Error Rate
- Open Issues Count

📌 Filters

- Site_Name
- Region
- Date

📈 Visualizations

- **Line Chart:** Reporting Compliance Trend Over Time
- **Bar Chart:** Site-wise Performance Score
- **Stacked Bar Chart:** Delay Distribution by Region
- **Donut Chart:** Issue Type Distribution
- **Scatter Plot:** Error Rate vs Delay Rate by Site

🧠 Insights Section

- Identified regions with **high delay rates**
- Highlighted **underperforming sites**
- Observed relationship between **delay and data errors**
- Provided actionable insights for improving reporting efficiency

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🛠️ Tools & Technologies Used

- Microsoft Excel (Data Cleaning & Preparation)
- Power BI (Data Modeling, DAX, Dashboard Design)
- DAX (KPI Calculations)

💡 Key Learnings

- Understanding of **real-world MIS reporting systems**
- Data cleaning and validation techniques
- Building **data models using star schema**
- Creating meaningful KPIs for business decisions
- Designing professional dashboards with storytelling

🚀 Conclusion

This project demonstrates the ability to transform raw operational data into a structured MIS reporting system. The dashboard enables management to monitor performance, identify inefficiencies, and take data-driven actions to improve overall operations.